

QA Testing for HR Analytics Dashboard

1. Total employees

```
SELECT COUNT(*) AS Total_Employees  
FROM dbo.HRData;
```

	Total_Employees
1	1470

2. Total attrition

```
SELECT SUM(CAST(Attrition AS INT)) AS Total_Attrition  
FROM dbo.HRData;
```

	Total_Attrition
1	237

3. Active employees

```
SELECT COUNT(*) - SUM(CAST(Attrition AS INT)) AS Active_Employees  
FROM dbo.HRData;
```

	Active_Employees
	1233

4. Attrition rate

```
SELECT  
    CAST(SUM(CAST(Attrition AS INT)) * 100.0 / COUNT(*) AS DECIMAL(10,2)) AS  
Attrition_Rate  
FROM dbo.HRData;
```

	Attrition_Rate
	16.12

5. Average age

```
SELECT CAST(ROUND(AVG(CAST(Age AS FLOAT)), 0) AS INT) AS Avg_Age  
FROM dbo.HRData;
```

	Avg_Age
	37

6. Attrition by gender

```
SELECT
    Gender,
    SUM(CAST(Attrition AS INT)) AS Attrition_Count
FROM dbo.HRData
GROUP BY Gender;
```

Gender	Attrition_Count
Male	150
Female	87

7. Attrition by department

```
SELECT
    Department,
    SUM(CAST(Attrition AS INT)) AS Attrition_Count
FROM dbo.HRData
GROUP BY Department
ORDER BY Attrition_Count DESC;
```

Department	Attrition_Count
R&D	133
Sales	92
HR	12

8. Attrition by education field

```
SELECT
    EducationField,
    SUM(CAST(Attrition AS INT)) AS Attrition_Count
FROM dbo.HRData
GROUP BY EducationField
ORDER BY Attrition_Count DESC;
```

	education_field	Attrition_Count
1	Life Sciences	89
2	Medical	63
3	Marketing	35
4	Technical Degree	32
5	Other	11
6	Human Resources	7